

Electromagnetic Compatibility:

CABLE SELECTION AND ROUTING



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It is frequently observed, and more so with those who design and manufacture electronic control panels, that the overall system often fails to pass electromagnetic compatibility (EMC) tests (particularly emissions), even though the manufacturer has meticulously taken all precautions to ensure product compliance, just because they forgot one crucial aspect of EMC design—cable routing and selection. And as the old saying goes, “All is well that ends well,” proper cable selection and routing is the last major design consideration to make a system complaint to EMC norms.

Cable coupling mechanisms

Three mechanisms exist by which EMI can be coupled on to cables. First, field-to-cable common-mode coupling occurs when wires or traces are routed away from the ground, forming a loop that acts as an antenna, causing common-mode EMI to be coupled into or out of the system.

Second, field-to-cable differential-mode coupling occurs when forward and return wires (or traces) are routed at a large distance from each other, forming a loop that acts as an antenna, causing differential EMI to be coupled into or out of the system.

Third, cable-to-cable coupling (cross-talk) is one where high-frequency signals on one cable can electrically couple (or talk) to adjacent cable through

stray capacitance formed between the two cables.

Cable classes

In order to properly segregate cables, these can be split into six classes, depending upon their application, voltage or current levels, frequency of signals these carry, types of signals these carry (like analogue or digital), components these connect to and so on. These classes are as follows:

Class 1. These are cables that carry highly-sensitive signals like low-level analogue signal (in mV) from transducers, instrument lines, radio receiver antenna cables, and cables carrying high-frequency digital signals such as Ethernet. Such cables are highly susceptible to noise and interference.

Class 2. These carry slightly-sensitive signals that are somewhat susceptible such as ordinary analogue signals under 1MHz and of the order of 1-10V and 4-20mA. These also include low-frequency digital signals like RS232 or RS485 and digital inputs/outputs (I/Os) like those from limit switches, encoders and control signals.

Class 3. These are cables carrying the noise source that is slightly interfering such as low-voltage AC distribution less than 1kV or DC power, linear power cables, control cables for inductive loads (such as relay coils, motor brakes and contactors) where proper transient suppression techniques like RC snubbers have been employed to control transients at the load. These also include motor cables from output of inverter drives but fitted with output filters as per manufacturer's specifications.

Class 4. These are cables carrying signals from strong noise sources that are highly interfering. These include output cables from frequency converters, inputs

Fig. 27:
Cable classes
(dimensions in
millimetres)

